

Split antecedents across conjunctions

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This study proposes and defends *dependency-building dynamic conjunction*, which tracks discourse dependencies established within its conjuncts. The proposal is motivated by novel empirical data from English and Japanese, in which the antecedents of a pronoun are spread across conjuncts. The resolution of the anaphora requires keeping track of the discourse dependencies established in the conjuncts as well as internal and external dynamicity. the dependency-building is a characteristic of the universal quantifier. This study suggests that conjunction should (at least sometimes) model dependencies as the universal quantifier does, and that the conjunction and universal quantifier are closer to each other than conventionally thought in the literature on dynamic semantics.

1 Introduction and summary

This paper addresses issues raised by the class of anaphoric dependencies observed in English (1) and Japanese (2), and proposes a dynamic resolution of them.¹

- (1) a. A boy saw a donkey, and a girl saw a monkey.
b. Each of them caught { *it* / *that animal* }.
 ↪ The boy caught the donkey he saw and the girl caught the monkey she saw.
- (2) a. *Otokonoko-ga roba-o mi-te, on'nanoko-ga saru-o mi-ta.*
 boy-NOM donkey-ACC see-and girl-NOM monkey-ACC see-PAST
 ‘A boy saw a donkey, and a girl saw a monkey.’
b. *Dochira-mo { sore-o / sono doobutsu-o } tsukamae-ta.*
 Each-all it-ACC that animal-ACC catch-PAST
 ‘Each caught { it / that animal }.’
 ↪ The boy caught the donkey he saw and the girl caught the monkey she saw.

I call the anaphoric relation in (1) and (2) *split antecedents across conjunction* (SAAC).² I contend that SAAC closely resembles *quantificational subordination* (Karttunen 1976, among others) exemplified in (3), and that SAAC is resolved in the same manner as quantificational subordinations. The quantifiers in the (b)-sentences of (1)/(2) do nothing more than *each (of them)* does in (3b): they induce *distributive interpretation*, which is described shortly below. What is unique about SAAC is how the *conjunction* manipulates the discourse.

¹The issue has been discussed in Charlow (2019, 6–7) with a DP conjunction, as well as Yagi (2023), in which I proposed a different (and not formally worked-out) direction from that proposed here, and Li (2025, 334), in which the observation is attributed to Simon Charlow and one of the anonymous reviewers.

²While the Japanese example in (2) is widely accepted, the acceptance rate of the English counterpart in (1) exhibits speaker variation. Nonetheless, speakers who find (1) degraded still accept the other parallel examples presented in Section 2. I have nothing to say about this inter-speaker variation. The robust acceptance of the Japanese data shows that the phenomenon is real and (1) might be degraded for an independent reason.

- (3) a. [Every student]⁰ wrote [a paper]¹.
 b. Each of them₀ submitted it₁ to *Journal of Semantics*.

Intuitively, sentence (3a) sets up pairs, or *dependencies*, of students and papers, where each student is paired with the paper they wrote. Subordinating sentence (3b) quantifies over those pairs, uttering that, for each pair, it is the case that the student submitted the paper to *Journal of Semantics*. The pronoun *it* therefore refers to the paper in the pair.

This intuition is formalized using a pluralized dynamic system (van den Berg, 1996; Nouwen, 2003, 2007; Brasoveanu, 2007), where sentence (3a) produces the *information state* in (4).

- (4) Information state for (3a)

F	0	1	...
f_1	student ₁	paper ₁	...
f_2	student ₂	paper ₂	...
$\vdots$	$\vdots$	$\vdots$	
f_n	student _{n}	paper _{n}	...

Information states are modeled as sets of *stacks* (a list of individuals).³ Given the stack f , we use $f(n)$ to retrieve the individual in the n th position of f . For example, F in (4) is a set containing stacks f_1 and f_2 such that $f_1(1) = \mathbf{paper}_1$, $f_2(0) = \mathbf{student}_2$. The information state captures the *dependency* between the students and paper: For each $f \in F$, $f(1)$ is the paper written by student $f(0)$.

Sentence (3b) subordinates to the dependency modeled in F . The intended reading is obtained by interpreting the scope of the quantifier *each (of them)* distributively against the information state F : The sentence asserts that for each $f \in F$, $f(0)$ submitted $f(1)$ to *Journal of Semantics*. This process is graphically presented in (5).

- (5) Distributive interpretation for (3b)

F	0	1	
f_1	student ₁	paper ₁	$\leftarrow 0$ submitted <i>it</i> ₁ to <i>Journal of Semantics</i>
f_2	student ₂	paper ₂	$\leftarrow 0$ submitted <i>it</i> ₁ to <i>Journal of Semantics</i>
$\vdots$	$\vdots$	$\vdots$	$\vdots$
f_n	student _{n}	paper _{n}	$\leftarrow 0$ submitted <i>it</i> ₁ to <i>Journal of Semantics</i>

The goal of the current paper is to analyze SAAC in parallel with quantificational subordination. To achieve this, the conjunctions in (1a)/(2a) should model the dependency between the boy/girl and the animal they see.

This direction requires us to depart from the standard dynamic analysis of conjunction as a sequential update of contexts (e.g., Heim, 1982; Kamp, 1981; Groenendijk and Stokhof, 1991). For our target sentences (1a)/(2a), the sequential dynamic conjunction first produces the information state G in (6) by processing the first conjunct, and then H by processing

³The use of stacks, instead of the more widely adopted *assignment functions*, is due to a mechanical but significant reason. The setup with stacks facilitates the important operations used in the proposal, namely *concatenation* and *forward popping*, as defined in Section 4.

the second. ($\star$ in F indicates that $f_1(0)$ and $f_1(1)$ are *empty*, which will be technically defined later in Section 3.)

$$(6) \quad \frac{F \mid \begin{array}{ccc} 0 & 1 & \dots \\ f_1 & \star & \star & \dots \end{array}}{\begin{array}{c} \xrightarrow{\text{A boy saw a donkey}} \\ G \mid \begin{array}{ccc} 0 & 1 & \dots \\ g_1 & \text{boy} & \text{donkey} & \dots \end{array} \end{array}}$$

$$\frac{\begin{array}{c} \xrightarrow{\text{A girl saw a monkey}} \\ H \mid \begin{array}{ccccc} 0 & 1 & 2 & 3 & \dots \\ h_1 & \text{boy} & \text{donkey} & \text{girl} & \text{monkey} & \dots \end{array} \end{array}}{\quad}$$

The information state H does not model the required dependency. What is needed instead is the *dependency-building conjunction*, which produces the information structure in (7). The distributive interpretation with respect to I in (7) resolves the pronoun *it* in SAAC and results in the intended interpretation.

$$(7) \quad \frac{I \mid \begin{array}{cc} 0 & 1 \\ i_1 & \text{boy} \quad \text{donkey} \\ i_2 & \text{girl} \quad \text{monkey} \end{array}}{\begin{array}{l} \leftarrow 0 \text{ caught } it_1 \\ \leftarrow 0 \text{ caught } it_1 \end{array}}$$

However, (7) faces technical and empirical issues: it is not dynamic for singular pronouns (in the sense of Groenendijk and Stokhof, 1991), as discussed in Section 4. Therefore, I propose a definition of a dependency-building dynamic conjunction that produces the information state in (8).

$$(8) \quad \frac{J \mid \begin{array}{cccccc} 0 & 1 & 2 & 3 & 4 & 5 \\ j_1 & \text{boy} & \text{donkey} & \text{girl} & \text{monkey} & \text{boy} & \text{donkey} \\ j_2 & \text{boy} & \text{donkey} & \text{girl} & \text{monkey} & \text{girl} & \text{monkey} \end{array}}{\quad}$$

J in (8) is obtained by concatenating I in (7) and H in (6). J preserves both the dependency built in I and the dynamicity built in H .

The process of obtaining J is informally described as follows. The sequential update remains the basic semantics of dynamic conjunction. The required dependency is obtained by extracting the difference between stacks. Consider again the sequential update in (6). The first conjunct of the target sentence, ‘*A boy saw a donkey*,’ updates F to G by adding the two discourse referents, **boy** and **donkey**, to the stack. These two discourse referents constitute the information that is newly added to G . That is, the difference between F and G is characterized by the new discourse referents in G , and this difference captures the dependency established by the first conjunct. Similarly, the difference between G and H is characterized by the newly added discourse referents **girl** and **monkey**, which, again, captures the dependency established by the second conjunct.

Thus, for two information states, F and G , we have the *transition* from F to G , represented as $\Delta^{F,G}$, which is a stack that characterizes the difference between F and G . (This operation is technically defined in Section 3 as the *forward popping* of stacks G/n , where n is the length of F .) For example, for F , G , H in (6), we obtain $\Delta^{F,G}$ and $\Delta^{G,H}$ as follows:

$$\begin{array}{l}
(9) \quad \frac{F}{f_1} \left| \begin{array}{ccc} 0 & 1 & \dots \\ \star & \star & \dots \end{array} \right. \\
\\
\begin{array}{c} \xrightarrow{\text{A boy saw a donkey}} \end{array} \quad \frac{G}{g_1} \left| \begin{array}{ccc} 0 & 1 & \dots \\ \text{boy} & \text{donkey} & \dots \end{array} \right. \\
\\
\frac{\Delta^{F,G}}{s_1} \left| \begin{array}{ccc} 0 & 1 & \dots \\ \text{boy} & \text{donkey} & \dots \end{array} \right. \\
\\
\begin{array}{c} \xrightarrow{\text{A girl saw a monkey}} \end{array} \quad \frac{H}{h_1} \left| \begin{array}{cccc} 0 & 1 & 2 & 3 & \dots \\ \text{boy} & \text{donkey} & \text{girl} & \text{monkey} & \dots \end{array} \right. \\
\\
\frac{\Delta^{G,H}}{s_2} \left| \begin{array}{ccc} 0 & 1 & \dots \\ \text{girl} & \text{monkey} & \dots \end{array} \right.
\end{array}$$

The union of $\Delta^{F,G}$ and $\Delta^{G,H}$ is the information state equivalent to I in (7), which models the dependencies established in the conjuncts.

$$(10) \quad \frac{\Delta^{F,G} \cup \Delta^{G,H}}{\begin{array}{c} s_1 \\ s_2 \end{array}} \left| \begin{array}{cc} 0 & 1 \\ \text{boy} & \text{donkey} \\ \text{girl} & \text{monkey} \end{array} \right.$$

Concatenating the information state $\Delta^{F,G} \cup \Delta^{G,H}$ with H in (9) results in the desired information state J in (8), which preserves both dynamicity and dependency.

Thus, the dependency-building dynamic conjunction, represented as $\wedge_{\forall}$ is defined as (11), where $*$ is the concatenation operation defined in Section 3. The update of F to H then to K is the familiar sequential dynamic conjunction. Dependency-building is achieved through the transitions $\Delta^{F,H}$ and $\Delta^{H,K}$.

$$(11) \quad F[\phi \wedge_{\forall} \psi]G \quad \text{iff} \quad \exists H, K : F[\phi]H \ \& \ H[\psi]K \ \& \ G = K * (\Delta^{F,H} \cup \Delta^{H,K})$$

The empirical contribution of this study is to analyze SAAC with a dependency-building dynamic conjunction that keeps track of both dynamicity and dependency. In addition, this study makes a theoretical commitment on the correspondence between conjunction and universal quantification. As shown in the brief illustration of the analysis of quantificational subordination, the dependency-building is a characteristic of the universal quantifier. The felicity of SAAC suggests that conjunction should (at least sometimes) model dependencies as the universal quantifier does, and that the conjunction and universal quantifier are closer to each other than conventionally thought in the literature on dynamic semantics. Thus, this study provides new empirical evidence for the theoretical resemblance of universal quantification and conjunction in dynamic semantics, where these two operations typically differ in significant respects (unlike other theories, which can define universal quantification as a generalized conjunction).

The remainder of this paper is organized as follows. Section 2 examines SAAC in more detail. Crucially, it will be shown that SAAC should not be analyzed as an instance of *paycheck* pronouns. Section 3 presents the necessary background for the proposed analysis by describing the analysis of quantificational subordination. Section 4 presents the proposal, followed by discussion on a few remaining issues in section (5). Section 6 compares the proposal with previous related works, and section 7 concludes the paper.

2 Observations

Robustness of SAAC

SAAC is cross-categorically robust. It can be replicated with temporal and locational pronouns.

- (12) a. A boy was in the park at five and a girl was in the station at six.
 b. Each of them got a phone call { *then* / *at that time* }.
 ~> The boy got a phone call at five, and the girl got a phone call at six.
- (13) a. *Otokonoko-ga go-ji-ni kooen-ni i-te, on'nanoko-ga roku-ji-ni*
 boy-NOM five-o'clock-DAT park-DAT be-and, girl-NOM six-o'clock-DAT
eki-ni i-ta.
 station-DAT be-PAST
 'A boy was in the park at five, and a girl was in the station at six.'
 b. *Dochira-mo sono toki denwa-o tot-ta.*
 Each-all that time call-ACC take-PAST
 'Each got a phone call then.'
 ~> The boy got a phone call at five, and the girl got a phone call at six.
- (14) a. A boy was in the park and a girl was in the station.
 b. Each of them got a phone call *there*.
 ~> The boy got a phone call in the park, and the girl got a phone call in the station.
- (15) a. *Otokonoko-ga go-ji-ni kooen-ni i-te, on'nanoko-ga roku-ji-ni*
 boy-NOM five-o'clock-DAT park-DAT be-and, girl-NOM six-o'clock-DAT
eki-ni i-ta.
 station-DAT be-PAST
 'A boy was in the park at five, and a girl was in the station at six.'
 b. *Dochira-mo soko-de denwa-o tot-ta.*
 Each-all there-DAT call-ACC take-PAST
 'Each got a phone call there.'
 ~> The boy got a phone call at the park, and the girl got a phone call at the station.

Dependency on a quantifier

The reading in question hinges on the presence of a quantifier. Minimal variants of (1)/(2) with a non-quantificational subject are degraded with the intended construal:⁴

⁴The quantificational force of the licensing quantifier need not be universal.

- (i) a. A boy saw a donkey and a girl saw a monkey.
 b. One of them caught { *it* / *that animal* }.
 ~> The boy caught the donkey he saw, and the girl caught the monkey she saw.

- (16) a. A boy saw a donkey, and a girl saw a monkey.
 b. They caught *it*.
 ?? $\rightsquigarrow$ The boy caught the donkey he saw, and the girl caught the monkey she saw.
- (17) a. *Otokonoko-ga roba-o mi-te, on'nanoko-ga saru-o mi-ta.*
 boy-NOM donkey-ACC see-and girl-NOM monkey-ACC see-PAST
 'A boy saw a donkey and a girl saw a monkey.'
 b. *Karera-wa { sore-o / sono doobutsu-o } tsukamaeta.*
 They-TOP it-ACC that animal-ACC caught
 'They caught { it / that animal }.'
 ?? $\rightsquigarrow$ The boy caught the donkey he saw, and the girl caught the monkey she saw.

To support the parallelism between SAAC and quantificational subordination that I argue for, note that quantificational subordination also hinges on the quantifier. The intended reading of (3) is absent (or considerably less robust) without *each*. (See, however, the discussion in section 5).

- (18) a. [Every student]⁰ wrote [a paper]¹.
 b. They₀ submitted it₁ to *Journal of Semantics*.

The absence of 'non-respective' reading

The (a)-sentences of (1)/(2) pair the boy with the donkey and the girl with the monkey. The (b)-sentences, intuitively, universally quantify over the pairs and assert that the relation **caught** also holds for each pair. Any theory must offer a way to correctly pair a person and an animal: A theory must prevent an unattested 'non-respective' reading, in which the persons and the animals are paired wrongly. For example, in (19) and (20), pairing Bill with the monkey and Mary with the donkey is not possible, even with the help of the context.

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- (ii) a. *Otokonoko-ga roba-o mi-te, on'nanoko-ga saru-o mi-ta.*
 boy-NOM donkey-ACC see-and girl-NOM monkey-ACC see-PAST
 'A boy saw a donkey and a girl saw a monkey.'
 b. *Dochira-ka-ga { sore-o / sono doobutsu-o } tsukamaeta.*
 which-some-NOM it-ACC that animal-ACC caught
 'One (of them) caught { it / that animal }.'
 $\rightsquigarrow$ The boy caught the donkey he saw or the girl caught the monkey she saw.
- (iii) a. A boy saw a donkey and a girl saw a monkey.
 b. Neither of them caught { *it* / *that animal* }.
 $\rightsquigarrow$ The boy did not catch the donkey he saw and the girl did not catch the monkey she saw.

- (19) (Bill and Mary have a deal: one has to catch the animal the other found. Then,)
- a. Bill found a donkey and Mary found a monkey.
 - b. Each of them caught *it*.
 $\nrightarrow$ Bill caught the monkey (Mary found) and Mary caught the donkey (Bill found).
- (20) a. Bill sold a donkey and Mary sold a monkey.
b. Each of them bought *it*.
 $\nrightarrow$ Bill bought the monkey (Mary sold) and Mary bought the donkey (Bill sold).

The absence of ‘non-respective’ reading suggests that the dependency in SAAC is formally established by conjunction rather than pragmatic reasoning.

Difference from paycheck pronoun

One possibility for resolving the SAAC is to interpret the pronoun as (*one of*) *the donkey or monkey that meets some property P*. In this case, *P* is presumably the property of being seen by an agent of the sentence, e.g., *seen by x*, where *x* would be bound by the quantifier. Such analyses have been implemented through the decomposition of pronouns (*E*-type analysis; Cooper 1979; Heim 1990; Elbourne 2001, 2005, among many others) or letting pronouns be more complicated than the simple individual variable (Gardent 1991; Hardt 1999; Keshet 2018).⁵ These analyses are primarily motivated by a *paycheck pronoun* (Karttunen 1969; Cooper 1979) exemplified in (21).

- (21) Every boy but Bill gave his paycheck to his mom. Bill gave *it* to his girlfriend.
 $\rightsquigarrow$ Bill gave *his paycheck* to his girlfriend.

This study does not attempt to refute any of those analyses. Nevertheless, the resolution of SAAC requires a different analysis from those for paycheck pronouns. This is because the intended interpretation of SAAC is felicitous with anaphoric expressions *that resist paycheck interpretation*. The Japanese pronoun *sore*, though felicitous in (2), is incompatible with the paycheck construal (Kurafuji 1998), as shown in (22). The same observation holds for the anaphoric demonstrative *sono+N*. *Sore* and *Sono+N* in (22b) cannot be interpreted as denoting John’s credit card.

- (22) a. *John igai-no dare-mo-ga jibun-no kurejittokaado-o tsuma-ni*
John other.than-GEN who-all-NOM self-GEN credit.card-ACC wife-DAT
watashita.
gave
‘Everyone but John gave his credit card to his wife.’
b. *John-wa { sore-o / sono-kurejittokaado-o } haha-ni watasita.*
John-TOP it-ACC that-credit.card-ACC mother-DAT gave
‘John gave *it/that credit card* to his mother.’
 $\rightsquigarrow$ #John gave John’s credit card to his mother.
(Kurafuji 1998: 130, modified, emphasis added.)

⁵Dynamic resolution of pronouns and the strategies for paycheck pronoun are not mutually incompatible, and cross-linguistic investigation has revealed that both are attested in natural language. See, for example, Kurafuji (1998), Schwarz (2009), and Patel-Grosz and Grosz (2017).

Any attempt to extend the strategy for the paycheck pronoun to (1)/(2) has to explain the contrast between the infelicitous (22) and the felicitous (2). Instead, I take the contrast as motivation to devise an analysis of (1)/(2) without the ‘paycheck’ strategy. For the absence of a paycheck construal for Japanese pronouns, I echo Kurafuji’s (1998) claim that the Japanese *sore* (and the anaphoric demonstrative *sono+N*) are analyzed as nothing more than an unstructured simple individual variable (with an overt restrictor in the case of demonstratives).⁶

Note that the anaphoric expressions *sore/sono+N* are felicitous in a configuration of quantificational subordination, as exemplified in (23). This observation further supports the similarity between quantificational subordination and SAAC.

- (23) a. *Dono-otokonoko-mo kujittokaado-o tsukutta.*
 Which-man-all credit.card-ACC made.
 ‘Every boy made a credit card.’
 b. *Doitsu-mo { sore-o / sono-kurejittokaado-o } haha-ni watasita.*
 Each.person-all it-ACC that-credit.card-ACC mother-DAT gave
 ‘Each (of them) gave *it/that credit card* to his mom.’
 (Subordination reading available)

Summary

Summarizing the above discussion, we have identified three desiderata for an analysis of SAAC.

- (24) a. SAAC hinges on the presence of a quantifier.
 b. SAAC only has the ‘respective’ reading. The ‘non-respective’ reading is not attested.
 c. SAAC is felicitous with an anaphoric expression that resists the paycheck construal.

3 Dynamics: An analysis of quantificational subordination

3.1 Stacks

Following plural dynamic theories (van den Berg 1996; Nouwen 2003, 2007; Brasoveanu 2007, 2011), this study assumes that sentences denote relations over information states. Information states are, in turn, modeled as sets of *stacks* (numbered lists of individuals; see Vermeulen 1993; van Eijck 2001; Nouwen 2003, 2007), graphically represented as follows:

(25)

F	0	1	2
f_1	a	b	c
f_2	d	e	f
f_3	g	h	i

⁶That is, *sore* and *sono+N* exhibit the properties of *Strong Definiteness* in the Schwarz’s (2009) dichotomy.

Each of $f_1 \dots f_3 \in F$ is a *stack*. For a number n , I say $f(n)$ is *defined* if and only if $f(n) \neq \star$, where $\star$ is the *dummy individual* that makes $P^m(\dots \star \dots)$ undefined for any P with arity m (Brasoveanu 2011).⁷ The length $|f|$ of stack f is $n + 1$, where n is the largest number n such that $f(n)$ is defined. For instance, the stack f in (26), has the length 3 ($= 2 + 1$).

$$(26) \quad \begin{array}{c|cccccc} & 0 & 1 & 2 & 3 & 4 & \dots \\ \hline f & a & b & c & \star & \star & \dots \end{array}$$

(27) The length of a stack f and an information state F

$$\begin{aligned} \text{a. } |f| &:= \begin{cases} 1 + \iota n : f(n) \neq \star \wedge f(l) = \star \text{ for all } l > n & \text{if } \exists n : f(n) \neq \star \wedge \\ & f(l) = \star \text{ for all } l > n \\ 0 & \text{if } \forall n : f(n) = \star \\ \text{undefined} & \text{otherwise} \end{cases} \\ \text{b. } |F| &:= \begin{cases} \iota n. \forall f \in F : |f| = n & \text{if } \forall f, f' \in F : |f| = |f'| \\ \text{undefined} & \text{otherwise} \end{cases} \end{aligned}$$

The other feature of the stack-based system is that it does not allow *destructive update*, that is, overwriting the existing information in the stack with new information. The introduction of a new referent is defined as (28). Intuitively, $f \rightarrow g$ appends a new referent at the end of the stack. (29) illustrates the update of f in (26).

(28) Stack update

$$f \rightarrow g \quad \text{iff} \quad \forall n < |f| : f(n) = g(n) \wedge \forall m > |f| : f(m) = g(m)$$

$$(29) \quad \begin{array}{c|cccccc} & 0 & 1 & 2 & 3 & 4 & \dots \\ \hline f & a & b & c & \star & \star & \dots \end{array} \xrightarrow{f \rightarrow g} \begin{array}{c|cccccc} & 0 & 1 & 2 & 3 & 4 & \dots \\ \hline g & a & b & c & d & \star & \dots \end{array}$$

These two features lead to straightforward definitions of *concatenation* and *forward popping* of stacks, which can be generalized to the operations on information states.⁸

(30) Concatenation

$$\begin{aligned} \text{a. } f * x &:= \iota g. f \rightarrow g \wedge g(|f|) = x && \text{(Concatenating stacks and individuals)} \\ \text{b. } f * g &:= f * g(0) * \dots * g(|g| - 1) && \text{(Concatenating stacks)} \\ \text{c. } F * G &:= \{f * g \mid f \in F \wedge g \in G\} && \text{(Concatenating information states)} \end{aligned}$$

(31) Forward Popping

$$\begin{aligned} \text{a. } f/n &:= \iota g. \forall m (0 \leq m) : g(m) = f(m + n) && \text{(Popping the first } n \text{ elements from a stack } f) \\ \text{b. } F/n &:= \{f/n \mid f \in F\} \end{aligned}$$

⁷Thus, I define an extended domain D^+ as $D^+ = D \cup \{\star\}$. Brasoveanu (2007; 2011) lets $\star$ be a universal falsifier to “keep the underlying logic classical” (Brasoveanu (2007): 215). Here, I would rather follow van den Berg (1996) and let $P^n(\dots \star \dots)$ be undefined, which, I suppose, better reflects our intuition. The utterance of *He₀ whisled* under the context in which no one has been mentioned in the discourse does not strike us *false*. That being said, nothing hinges on this technical choice.

⁸The definition of concatenation is based on Nouwen (2007). The operation of forward popping is new.

(32) and (33) respectively illustrate the two operations. Concentration of the stack has been assumed in the aforementioned literature. Forward popping is a new one, and it removes a certain amount of information from a stack. Intuitively, $f/2$ is the stack that results from deleting the first two individuals, $f(0)$ and $f(1)$, from the stack f .

$$(32) \quad \frac{F}{\begin{array}{c|ccc} & 0 & 1 & 2 \\ \hline f_1 & a & b & c \\ f_2 & d & e & f \end{array}} * \frac{G}{\begin{array}{c|cc} & 0 & 1 \\ \hline g_1 & g & h \end{array}} = \frac{H}{\begin{array}{c|ccccc} & 0 & 1 & 2 & 3 & 4 \\ \hline h_1 & a & b & c & g & h \\ h_2 & d & e & f & g & h \end{array}}$$

$$(33) \quad \frac{F}{\begin{array}{c|ccccc} & 0 & 1 & 2 & 3 & 4 \\ \hline f_1 & a & b & c & d & e \\ f_2 & f & g & h & i & j \end{array}} / 2 = \frac{G}{\begin{array}{c|ccc} & 0 & 1 & 2 \\ \hline g_1 & c & d & e \\ g_2 & h & i & j \end{array}}$$

3.2 Basic dynamics

To illustrate the dynamic effect, consider (34a), translated into (34b).⁹

- (34) a. $[A \text{ man}]^0$ bought $[a \text{ book}]^1$.
b. $\rightsquigarrow \epsilon_0 \wedge \mathbf{sg}(0) \wedge \mathbf{man}(0) \wedge \epsilon_1 \wedge \mathbf{sg}(1) \wedge \mathbf{book}(1) \wedge \mathbf{bought}(0, 1)$

The relevant definitions and assumptions are as follows.

- Sentences denote relations over pairs of information states.
- Predicates are lifted to take sets: an n -place predicate takes n -number of sets of individuals. (cf. van den Berg 1996)
- Object-language notations such as (34b) are subject to interpretation via interpretation function $\llbracket - \rrbracket$.

⁹A few nontrivial assumptions underlie the translation here. The use of 0 and 1 in the superscripts, given the definition of ϵ below, assumes that the incoming information states – the information states that the sentence is updating – has the length 0. The assumption is not necessary. Given the definition of stack update that allows appending a new individual only to the end of a stack, superscripts for natural language indefinites (e.g. 0 in $[A \text{ man}]^0$) could be eliminated, and the argument for the predicate $\mathbf{man}$ should rather be the ‘newest element of the incoming stack’, possibly represented by a negative number -1 . If we elect to do so, the subscripts for ϵ and the first conjunct of the definition ($n = |f|$) could also be dropped, leading to the translation in (i). Notice that -1 in $\mathbf{book}(-1)$ picks up the last element in the stack *after* processing the second occurrence of ϵ . This translation also lets us discard the assumption that the incoming information state is empty. See Nouwen (2003) for a compositional implementation of this line of translation.

- (i) a. A man bought a book.
b. $\rightsquigarrow \epsilon \wedge \mathbf{sg}(-1) \wedge \mathbf{man}(-1) \wedge \epsilon \wedge \mathbf{sg}(-1) \wedge \mathbf{book}(-1) \wedge \mathbf{bought}(-2, -1)$

As far as the purpose of the present paper is concerned, however, the technical difference is trivial. In order to keep the transparent translation between the natural and object languages and the intuitive representation of anaphoricity, I sidestep this issue and assume that subscripts for ϵ are wisely chosen. Nothing empirical hinges on this choice.

- $F \llbracket R(n_1, \dots, n_m) \rrbracket G$ iff $F = G \wedge \langle F(n_1), \dots, F(n_m) \rangle \in I(R)$,
where $F(n) := \{f(n) \mid f \in F\}$
- $F \llbracket \epsilon_n \rrbracket G$ iff $n = |F|$ and $\forall f \in F : \exists g \in G : f \rightarrow g$
and $\forall g \in G : \exists f \in F : f \rightarrow g$
- $F \llbracket \phi \wedge \psi \rrbracket G$ iff $\exists H : F \llbracket \phi \rrbracket H \wedge H \llbracket \psi \rrbracket G$
- $F \llbracket \text{sg}(n) \rrbracket G$ iff $|F(n)| = 1$

The conjunction $\wedge$ is defined as the standard sequential dynamic conjunction. I later argue that the analysis of SAAC involves a different entry. For now, with these assumptions, (34b) updates an input information state as shown in (35). $G(1)$ and $G(2)$ store a singular man and a singular book, respectively, such that $\langle G(1), G(2) \rangle \in I(\text{bought})$, where I is the interpretation function. I assume that the singularity requirement comes from the singular inflections of the indefinites *a man* and *a book*, although nothing hinges on this assumption.

$$(35) \quad \frac{F \mid \begin{array}{ccc} 0 & 1 & \dots \\ f_1 & \star & \star \dots \end{array}}{\quad} \xrightarrow{(34b)} \frac{G \mid \begin{array}{ccc} 0 & 1 & \dots \\ g_1 & \text{man}_1 & \text{book}_2 \dots \end{array}}{\quad}$$

3.3 Universal quantification and quantificational subordination

The dynamic effect of the universally quantified sentence in (36) is twofold (supposing that the subject takes scope over the object).

$$(36) \quad [\text{Every farmer}]^0 \text{ owns } [\text{a donkey}]^1.$$

- The output information state G stores all the farmers under the index 0.
- For each farmer $g(0)$ ($g \in G$), $g(1)$ stores a donkey that $g(0)$ owns.

The first effect, storing *all* the (relevant) farmers, is induced by the *maximization operator* (max^n). In order to model the second effect, it is necessary to ‘zoom in’ on each $g \in G$ so that the appropriate farmer-donkey relations are stored in $g(0)$ and $g(1)$ for all $g \in G$. This effect is formalized using the *distribution operator* (dist_n).¹⁰

- $F \llbracket \text{max}^n(\phi) \rrbracket G$ iff $F \llbracket \epsilon_n \wedge \phi \rrbracket G \wedge \forall H : F \llbracket \epsilon_n \wedge \phi \rrbracket H \rightarrow H(n) \subseteq G(n)$
- $F \llbracket \text{dist}_n(\phi) \rrbracket G$ iff $F(n) = G(n) \wedge \forall x \in F(n) : F|_{n=x} \llbracket \phi \rrbracket G|_{n=x}$,
where $F|_{n=x} := \{f \in F : f(n) = x\}$

Universally quantified sentences are translated as in (37), where P and Q are translated from NP and VP , respectively. For illustration, (36) is analyzed as (38).

$$(37) \quad \text{Every}^n \text{ } NP \text{ } VP \\ \rightsquigarrow \text{max}^n(P(n)) \wedge \text{dist}_n(Q(n))$$

¹⁰The two definitions here are borrowed from Brasoveanu (2011).

$$(38) \quad \max^0(\text{farmer}(0)) \wedge \text{dist}_0(\epsilon_1 \wedge \text{donkey}(1) \wedge \text{sg}(1) \wedge \text{own}(0, 1))$$

Suppose that we have three farmers, **farmer**₁, **farmer**₂, and **farmer**₃ who own donkeys **donkey**₁, **donkey**₂, and **donkey**₃, respectively. The update proceeds as follows.

$$(39) \quad \begin{array}{c|ccc} F & 0 & 1 & 2 \\ \hline f_1 & \dots & \dots & \dots \end{array} \xrightarrow{\max^0(\text{farmer}(0))} \begin{array}{c|ccc} H & 0 & 1 & 2 \\ \hline h_1 & \text{farmer}_1 & \dots & \dots \\ h_2 & \text{farmer}_2 & \dots & \dots \\ h_3 & \text{farmer}_3 & \dots & \dots \end{array}$$

$$\xrightarrow{\text{dist}_0(\epsilon_1 \wedge \text{donkey}(1) \wedge \text{sg}(1) \wedge \text{own}(0, 1))} \begin{array}{c|ccc} G & 0 & 1 & 2 \\ \hline g_1 & \text{farmer}_1 & \text{donkey}_1 & \dots \\ g_2 & \text{farmer}_2 & \text{donkey}_2 & \dots \\ g_3 & \text{farmer}_3 & \text{donkey}_3 & \dots \end{array}$$

The first conjunct of (38), namely $\max^0(\text{farmer}(0))$, requires all three farmers to be stored under index 0 (see H). The second conjunct then induces a distributive update: for each assignment function $h \in H$ (i.e., for each row, intuitively), it introduces a single donkey owned by farmer $h(0)$. The process results in G in the above figure, in which each row captures the farmer-donkey dependencies.

The farmer-donkey dependency stored in the information state is exploited in the analysis of quantificational subordination. Suppose that (39) is followed by the utterance in (40). With the information state G in (39), (40) states that each farmer stored in G above takes care of the donkey s/he owns.

$$(40) \quad \text{Each of them}_0 \text{ takes care of it}_1 \text{ well.}$$

To implement intuition formally, (40) is analyzed as (41). The quantifier *each (of them)*₀ is an overt realization of the distributive operator dist_0 . Given G in (39), (41) distributively checks each $g \in G$ in terms of whether $g(0)$ takes care of $g(1)$ well and $g(0)$ is singular. The singularity requirement again comes from the singular morphology of the pronoun *it*.

$$(41) \quad \text{dist}_0(\text{take_care_of_well}(0, 1) \wedge \text{sg}(1))$$

$$(42) \quad \begin{array}{c|ccc} F & 0 & 1 & 2 \\ \hline f_1 & \dots & \dots & \dots \end{array} \xrightarrow{(38)} \begin{array}{c|cc} G & 0 & 1 \\ \hline g_1 & \text{farmer}_1 & \text{donkey}_1 \\ g_2 & \text{farmer}_2 & \text{donkey}_2 \\ g_3 & \text{farmer}_3 & \text{donkey}_3 \end{array} \begin{array}{l} \leftarrow \text{take_care_of_well}(0, 1) \wedge \text{sg}(1) \\ \leftarrow \text{take_care_of_well}(0, 1) \wedge \text{sg}(1) \\ \leftarrow \text{take_care_of_well}(0, 1) \wedge \text{sg}(1) \end{array}$$

The distributive interpretation achieves two things. Firstly, it allows the subordinated quantification to quantify over the farmer-donkey dependencies stored in G . This is the formal reflex of the intuition that (41) quantifies over pairs of farmers and donkeys mentioned in the previous discourse. Secondly, it relaxes the singularity requirement imposed by the

singular pronoun *it*. The requirement is satisfied because it is evaluated against $G|_{0=x}$ for each $x \in G(0)$: $G|_{0=x}(1)$ stores only one donkey.¹¹

The subordinating reading should, therefore, be ruled out without the distributive operator. It is for this reason that the corresponding sentences without *each*, (44), are degraded in the intended reading (but see also the discussion in section 5). Focusing on the resolution of *it*, the sentences in (44) require that $G(0)$ be a singleton set. This is satisfied if, in (38), *a donkey* takes scope over the universal quantification or if a donkey does not co-vary with farmers. Either way, the intended reading—farmers own different donkeys and each farmer takes care of the donkey s/he owns—is not obtained.

(44) (Uttered after (38))

- a. He takes care of it₁ well.
- b. They take care of it₁ well.

At this point, the theoretical problems caused by SAAC have become clear. Focusing on the English example, we consider the possible indexations in (45). With *and* being analyzed as the sequential conjunction $\wedge$, the sentence would produce the information state in (46).

- (45) a. [A boy]⁰ saw [a donkey]¹, and [a girl]² saw [a monkey]³.
 b. $(\epsilon_0 \wedge \mathbf{sg}(0) \wedge \mathbf{boy}(0) \wedge \epsilon_1 \wedge \mathbf{sg}(1) \wedge \mathbf{donkey}(1) \wedge \mathbf{saw}(0, 1)) \wedge$
 $(\epsilon_2 \wedge \mathbf{sg}(2) \wedge \mathbf{girl}(2) \wedge \epsilon_3 \wedge \mathbf{sg}(3) \wedge \mathbf{monkey}(3) \wedge \mathbf{saw}(2, 3))$

(46)

F	0	1	2	3
f_1	boy	donkey	girl	monkey

This information state is not suitable for the distributive interpretation by **dist**, because it does not preserve the *seeing*-dependency between the individuals and the animals. More intuitively, a problem arises because the two individuals and the two animals are stored under different indices. For example, consider $\mathbf{dist}_0(\mathbf{caught}(0, ??))$, where ?? is a placeholder for the index carried by the pronoun. Whatever the index in ??, the resultant reading is that where the man caught one specific animal, i.e., the donkey or the monkey.

¹¹Recall that sentences denote *relations* over pairs of information states. If **farmer**₁ owns two donkeys, for example, the update denotes a relation $\{\langle F, G_1 \rangle, \langle F, G_2 \rangle\}$ where G_1 is exactly like G in (39) and G_2 differs from G only in that $g_1(1) = \mathbf{donkey}_8$. The subordinating sentence tests if G_1 and G_2 meet the condition specified in (41). Given an information state F , let $\llbracket \phi \rrbracket$ be true if and only if $F \llbracket \phi \rrbracket G$ for some G (following Groenendijk and Stokhof 1991). Then the discourse is true if at least one of G_1/G_2 meets the condition in (41), resulting in an analogue of *weak reading* of donkey anaphora. Whether or not this prediction is valid is an empirical matter, and I am not aware of an explicit discussion on the strong/weak reading of quantificational subordination, except that Brasoveanu (2007:237) notes that discourses like (36)/(40) have a strong *uniqueness* intuition that each farmer owns exactly one donkey. In order to impose such a reading, Brasoveanu (2007) suggests translating (36) as (i), in which the singularity requirement is replaced with the **max** operator, by which *all* the donkeys owned by each farmer are introduced. The singular requirement of a pronoun then requires that each farmer owns exactly one donkey, as desired.

- (43) $\mathbf{max}^0(\mathbf{farmer}(0)) \wedge \mathbf{dist}_0(\mathbf{max}^1(\mathbf{donkey}(1) \wedge \mathbf{own}(0, 1)))$

Nevertheless, the original translation in (38) should also be available to derive a weak reading of donkey anaphora, and if so, nothing precludes quantificational subordination having a weak reading as well. I will leave this matter open for now.

Therefore, the analysis of SAAC calls for a *dependency-preserving conjunction*, which formally characterizes the *seeing* dependency established in the (a)-sentences. The next section is devoted to formally defining the necessary conjunction.

4 Analysis

In this section, I argue that (1)/(2), repeated below, should be analyzed in a manner similar to that sketched above for quantificational subordination.

- (1) a. A boy saw [a donkey], and a girl saw [a monkey].
 b. Each of them caught { *it* / *that animal* }.
 $\rightsquigarrow$ The boy caught the donkey and the girl caught the monkey.
- (2) a. *Otokonoko-ga roba-o mi-te, on'nanoko-ga saru-o mi-ta.*
 boy-NOM donkey-ACC see-and girl-NOM monkey-ACC see-PAST
 ‘A boy saw a donkey and a girl saw a monkey.’
 b. *Dochira-mo { sore-o / sono doobutsu-o } tsukamaeta.*
 Each-all it-ACC that animal-ACC caught
 ‘Each caught { *it* / *that animal* }.’
 $\rightsquigarrow$ The boy caught the donkey he saw, and the girl caught the monkey she saw.

In doing so, I keep the analysis of the (b)-sentences analogous to (41), repeated below in (47): The quantifiers *each/dochiramo* induce distributive interpretation, and the anaphoric expressions *it/sore/sono doobutsu* are translated into a simple variable.

$$(47) \text{ dist}_0(\text{take_care_of_well}(0,1) \wedge \text{sg}(1))$$

What is new in the analysis will be the analysis of conjunction. The conjunction in SAAC should keep track of the person-animal dependency (i.e., the dependency via the *seeing* relation) encoded in the (a)-sentences of (1)/(2) as well as the dynamics of conjunction. Therefore, the goal is to analyze the (a)-sentences so that they produce the information structure F depicted in (48), which will be an input for the distributive interpretation of the (b)-sentences.

J	0	1	2	3	4	5	
(48) j_1	boy	donkey	girl	monkey	boy	donkey	$\leftarrow 4 \text{ caught } it_5$
j_2	boy	donkey	girl	monkey	girl	monkey	$\leftarrow 4 \text{ caught } it_5$

4.1 Dependency-Building Conjunction (Non-Dynamic)

To motivate this proposal, I begin with a definition of a dependency-building conjunction that does not work well. To avoid confusion, the standard dynamic conjunction assumed thus far is denoted as $\wedge$ and the new proposed entry is denoted as $\wedge_{\forall}$, reflecting the intuition that it has a dynamic effect similar to universal quantification.

- (49) Dependency-building conjunction (internally and externally static)
 $F[\phi \wedge_{\forall} \psi]G \text{ iff } \exists H, K : F[\phi]H \wedge F[\psi]K \wedge G = H \cup K$

This entry evaluates the conjuncts independently and summarizes the results.¹² Suppose that the (a)-sentences of (1)/(2) are analyzed using the indexation in (50a), which is translated into (50b). Its dynamic effect is depicted in (51).

- (50) a. [A boy]⁰ saw [a donkey]¹, and [a girl]⁰ saw [a monkey]¹.
 b. $(\boxed{A} \epsilon_0 \wedge \mathbf{sg}(0) \wedge \mathbf{boy}(0) \wedge \epsilon_1 \wedge \mathbf{sg}(1) \wedge \mathbf{donkey}(1) \wedge \mathbf{saw}(0, 1)) \wedge_{\forall}$
 $(\boxed{B} \epsilon_0 \wedge \mathbf{sg}(0) \wedge \mathbf{girl}(0) \wedge \epsilon_1 \wedge \mathbf{sg}(1) \wedge \mathbf{monkey}(1) \wedge \mathbf{saw}(0, 1))$

$$(51) \quad \frac{F \mid \begin{array}{ccc} 0 & 1 & \dots \\ f & \dots & \dots \end{array}}{\begin{array}{c} \boxed{A} \\ \hline \end{array}} \xrightarrow{\quad} \frac{H \mid \begin{array}{ccc} 0 & 1 & \dots \\ h_1 & \mathbf{boy} & \mathbf{donkey} \end{array}}{\begin{array}{c} \boxed{B} \\ \hline \end{array}} \xrightarrow{\quad} \frac{G \mid \begin{array}{ccc} 0 & 1 & \dots \\ h_1 & \mathbf{boy} & \mathbf{donkey} \\ k_1 & \mathbf{girl} & \mathbf{monkey} \end{array}}{\begin{array}{c} \boxed{B} \\ \hline \end{array}} \xrightarrow{\cup} \frac{G \mid \begin{array}{ccc} 0 & 1 & \dots \\ h_1 & \mathbf{boy} & \mathbf{donkey} \\ k_1 & \mathbf{girl} & \mathbf{monkey} \end{array}}{\begin{array}{c} \boxed{B} \\ \hline \end{array}}$$

Each conjunct induces an independent update, and the result is summed up in G in (51). It does produce an information state that stores the person-animal dependencies. The distributive interpretation of G resolves the anaphora in (1b) and (2b).¹³

However, this version of the dependency-building conjunction loses the dynamicity of the usual conjunction: it is internally static, and does not support external dynamicity for singular pronouns. Given that the dynamicity has been taken as a core part of the semantics of conjunction, this is not a theoretically welcome consequence. Furthermore, empirical observations reveal that the conjunction for SAAC should be internally and externally dynamic. The internal dynamicity is shown by (52)/(53), in which the intended reading is not hindered by the internal anaphoricity.¹⁴

- (52) a. A man saw a donkey and his son saw a monkey.
 b. Each of them caught it.
 $\rightsquigarrow$ The man caught the donkey he saw and his son caught the monkey he (the

¹²Li (2025) proposed a conjunctive connective that is akin to (49) to analyze reference to dependencies established by (multiple) *wh*-questions, in which the connective collects the updates by the alternative propositions denoted by *wh*-questions. Therefore, the dynamicity between the connected clauses was not an issue there.

¹³One may wonder if the surface word order has a significant effect in the validity of this line of analysis, given that information states preserve the order of references introduced to them. For example, one might wonder if a minimal variant of (50a) that replaces the second conjunct by *a girl gave a carrot to a monkey* fails to establish the needed dependency, because then $G(1) = \{\mathbf{donkey}, \mathbf{carrot}\}$. This is not necessarily the case, however. Information states preserve the evaluation order (or the scopal relation) of the sentence, which does not necessarily correspond to the surface word order. Given the exceptionally flexible scope behavior of indefinites (Reinhart 1997, among many others), the sentence could be processed in a way that it establishes the necessary dependency.

¹⁴See also the discussion around (66) for a more complex observation on the internal dynamicity with three conjuncts.

son) saw.

- (53) a. *Otoko-ga roba-o mi-te, sono-musuko-ga saru-o mi-ta.*
 man-NOM donkey-ACC see-and the-son-NOM monkey-ACC see-PAST
 ‘A man saw a donkey and his son saw a monkey.’
 b. *Dochira-mo { sore-o / sono doobutsu-o } tsukamaeta.*
 each-all it-ACC that animal-ACC caught
 ‘One (of them) caught { it / that animal }.’
 $\rightsquigarrow$ The man caught the donkey he saw and his son caught the monkey he (the son) saw.

The problem regarding external dynamicity is highlighted by the following discourse.

- (54) a. [A man]⁰ saw [a donkey]¹, and [a girl]⁰ saw [a monkey]¹.
 b. Each of them₀ caught it₁ in front of her_? siblings,
 c. but [the donkey]_? escaped later from his _? place.

The dependency-building conjunction and the distributive interpretation resolve the anaphora for *it* in (54b). However, the resolution of the other *singular* pronouns and definite require external dynamicity of conjunction. The singular pronoun *her* is not resolved either because it is within the scope of distributive quantifier by which it has to be evaluated against h_1 ($h_1(0) = \text{boy}$) that does not provide the intended referent *girl*, or because it is outside the scope of distributive quantifier thereby the singularity requirement impeded the anaphoric resolution. The latter reason also impeded the resolution of *the donkey* and *his* in (54c). Consider the formula in (55), evaluated against G . The anaphoric definite *the donkey* is supposed to pick up the referent $G(?)$ such that $G(?)$ is a *singleton* set containing a donkey. However, neither 0 nor 1 satisfies this requirement: $G(0)$ and $G(1)$ are not singleton sets. Therefore, the definition in (49) makes an incorrect prediction that the discourse in (54) is infelicitous, or, more generally, that a singular anaphoric expression in the dependency-building conjunction is not supported.

- (55) $\text{donkey}(?) \wedge \text{escaped}(?) \wedge \text{sg}(?)$

4.2 Dependency-building conjunction (dynamic)

The discussion thus far has revealed that the full-fledged analysis of SAAC calls for modelling (i) the usual dynamicity of the conjunction and (ii) the dependency explicated in each conjunct simultaneously. To achieve these two desiderata, I define a dependency-building conjunction that produces the information state in (8).¹⁵

J	0	1	2	3	4	5	
(58) j_1	boy	donkey	girl	monkey	boy	donkey	$\leftarrow 4 \text{ caught } it_5$
j_2	boy	donkey	girl	monkey	girl	monkey	$\leftarrow 4 \text{ caught } it_5$

¹⁵The proposed analysis also gives a straightforward dynamic analysis of some cases of plural pronouns. Plural pronouns referring to separately introduced singular antecedents have been commonly analyzed with multiple indexation and (optionally) a corresponding rule of discourse update. For example, *they* in (i) is analyzed with double indexation $0 + 1$ and a corresponding rule that introduces to the second position of an information state a plural individual consisting of John and Mary.

The information state J in (58) resolves the anaphora in question while preserving internal and external dynamicity for singular anaphoric expressions. Intuitively, the 0th to 3rd positions of the stack encode dynamicity, while the 4th and 5th positions preserve the dependency. As an illustration, consider (59). The singular pronoun his_0 refers to the man stored in $J(0)$. The singularity requirement is satisfied, because $J(0)$ is a singleton set. The inanimate singular pronoun it_5 is resolved through the distributive interpretation of J on par with quantificational subordination. The two singular expressions in (59c) are resolved in the usual manner: they select the referent in $J(1)$ and $J(0)$. The singularity requirement is also met here, because both $J(0)$ and $J(1)$ are singleton sets.

- (59) a. [A man]⁰ saw [a donkey]¹, and [his_0 son]² saw [a monkey]³.
 b. Each of them₄ caught it₅,
 c. but [the donkey]₁ escaped later from his₀ place.

I now turn to formally defining the dependency-building conjunction that produces (59). I propose keeping the sequential update, the heart of the dynamicity of conjunction, as the basic semantics of conjunction. The necessary dependency is extracted from the sequential update. The definition is laid out in (60).

- (60) Dependency-preserving conjunction
 $F \llbracket \phi \wedge_{\forall} \psi \rrbracket G \text{ iff } \exists H, K : F \llbracket \phi \rrbracket H \ \& \ H \llbracket \psi \rrbracket K \ \& \ G = K * (\Delta^{F,H} \cup \Delta^{H,K})$

- (61) Transition $\Delta^{F,G}$ for information states F, G
 $\Delta^{F,G} := G / |F|$

The update of F to H then to K is the sequential update for the standard dynamic conjunction. It guarantees that the dynamics of the conjunction are preserved. The resultant information state of the sequential update, K , does not grasp the needed dependency. The dependency is established through the *transitions* $\Delta^{F,H}$ and $\Delta^{H,K}$. Intuitively, the transition $\Delta^{F,H}$ extracts the referents in H that are newly added when F is updated to H . In our case, the transitions $\Delta^{F,H}$ and $\Delta^{H,K}$ single out the pairs **boy-donkey** and **girl-monkey**, for which the *seeing* dependency holds.

The sentences in (1a/b) are now translated into (62a/b). The update is illustrated in (63). H is the result of a sequential dynamic update. $\Delta^{F,G}$ singles out the **boy-donkey** dependency, and $\Delta^{G,H}$ singles out the **girl-monkey** dependency. The union $\Delta^{F,G} \cup \Delta^{G,H}$

- (56) John⁰ plays the guitar and Mary¹ plays the piano. They₀₊₁² formed a band.

If the conjunction in (i) is analyzed as $\wedge_{\forall}$, however, the dependency-building conjunction provides an information state in which the plurality consisting of John and Mary has already been established, as described in (57). The analysis thus provides a straightforward analysis in which the plural pronoun *they* carries one anaphoric index, namely 2, to refer to the (discourse) plural individual. I thank Simon Charlow (p.c.) for pointing this out.

- (57)

G	0	1	2
g_1	j	m	j
g_2	j	m	m

is an information state that grasps these dependencies, which is concatenated with H . The resultant information state J is what we have aimed at; it preserves both the dynamicity and dependency. The anaphora in question is resolved through the distributive interpretation with respect to J .

$$(62) \quad \begin{array}{l} \text{a. } (\epsilon_0 \wedge \mathbf{sg}(0) \wedge \mathbf{boy}(0) \wedge \epsilon_1 \wedge \mathbf{sg}(1) \wedge \mathbf{donkey}(1) \wedge \mathbf{saw}(0, 1)) \wedge_{\forall} \\ \quad \epsilon_0 \wedge \mathbf{sg}(0) \wedge \mathbf{girl}(0) \wedge \epsilon_1 \wedge \mathbf{sg}(1) \wedge \mathbf{monkey}(1) \wedge \mathbf{saw}(0, 1)) \\ \text{b. } \mathbf{dist}_4(\mathbf{caught}(4, 5)) \end{array}$$

$$(63) \quad \frac{F \mid 0 \quad 1 \quad \dots}{f_1 \mid \star \quad \star \quad \dots}$$

$$\xrightarrow{\text{A boy saw a donkey}}$$

$$\frac{G \mid 0 \quad 1 \quad \dots}{g_1 \mid \mathbf{boy} \quad \mathbf{donkey} \quad \dots}$$

$$\frac{\Delta^{F,G} \mid 0 \quad 1 \quad \dots}{s_1 \mid \mathbf{boy} \quad \mathbf{donkey} \quad \dots}$$

$$\xrightarrow{\text{A girl saw a monkey}}$$

$$\frac{H \mid 0 \quad 1 \quad 2 \quad 3 \quad \dots}{h_1 \mid \mathbf{boy} \quad \mathbf{donkey} \quad \mathbf{girl} \quad \mathbf{monkey} \quad \dots}$$

$$\frac{\Delta^{G,H} \mid 0 \quad 1 \quad \dots}{s_2 \mid \mathbf{girl} \quad \mathbf{monkey} \quad \dots}$$

$$\xrightarrow{\Delta^{F,G} \cup \Delta^{G,H}}$$

$$\frac{\Delta^{F,G} \cup \Delta^{G,H} \mid 0 \quad 1 \quad \dots}{\begin{array}{l} s_1 \mid \mathbf{boy} \quad \mathbf{donkey} \quad \dots \\ s_2 \mid \mathbf{girl} \quad \mathbf{monkey} \quad \dots \end{array}}$$

$$\xrightarrow{H*(\Delta^{F,G} \cup \Delta^{G,H})} \frac{J \mid 0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5}{\begin{array}{l} j_1 \mid \mathbf{boy} \quad \mathbf{donkey} \quad \mathbf{girl} \quad \mathbf{monkey} \quad \mathbf{boy} \quad \mathbf{donkey} \quad \leftarrow 4 \text{ caught } it_5 \\ j_2 \mid \mathbf{boy} \quad \mathbf{donkey} \quad \mathbf{girl} \quad \mathbf{monkey} \quad \mathbf{girl} \quad \mathbf{monkey} \quad \leftarrow 4 \text{ caught } it_5 \end{array}}$$

The proposed analysis preserves the internal and external dynamicity. The information introduced in the first conjunction is available for the second conjunct, thanks to the sequential update. The sequential update also allows the referent introduced within the conjunction to be picked up in the following discourse. For example, *the donkey*₁ refers to the donkey seen and caught by the man. The singularity requirement is met because $J(1)$ is a singleton set.

Furthermore, the proposed analysis satisfies the three desiderata discussed in section 2, repeated below.

- SAAC hinges on the presence of a quantifier.
- SAAC only has the ‘respective’ reading. The ‘non-respective’ reading is not attested.
- SAAC is felicitous with anaphoric expressions that resist the paycheck construal.

The intended reading of SAAC hinges on the presence of a quantifier for the same reason quantificational subordination requires a quantifier: the singularity requirement must be relaxed by the distributive operator. More specifically, consider the utterance in (64):

- (64) a. [A man]⁰ saw [a donkey]¹, and [his₀ son]² saw [a monkey]³.
 b. He₄ caught it₅.
 $\rightsquigarrow \text{caught}(4, 5) \wedge \text{sg}(4) \wedge \text{sg}(5)$

(64b), interpreted against J in (63), is degraded with the intended interpretation (where the man caught the donkey and his son caught the monkey) because of the singularity requirement imposed by the singular pronouns *he* and *it*. Without a quantifier, the sentence is interpreted in a *non-distributive* manner. However, this requires that $|J(4)| = |J(5)| = 1$, which is not the case. Therefore, the intended interpretation necessitates a quantifier, or, more accurately, a distributive operator accompanying a quantifier, precisely for the same reason that quantificational subordination requires a quantifier.

The ‘non-respective’ reading is not available because each ‘row’ in an information state stores a dependency encoded in each conjunct: the man-donkey dependency encoded in the first conjunct is stored in the 4th and 5th positions of j_1 above, and the son-monkey dependency is in j_2 . The distributive operator is sensitive to such dependencies. Therefore, the resulting reading preserves them and does not allow non-respective pairings.

Lastly, the proposed analysis takes the singular expressions in (1)/(2) as a simple individual variable, not as any complicated entities such as a covertly decomposed ‘E-type’ expression. The fact that they are analyzed as a simple variable in the proposed resolution teases (1)/(2) apart from paycheck pronouns, and the contrast between the felicitous Japanese example in (2) and the infelicitous paycheck sentence in (21) comes as no surprise in view of Kurafuji’s (1998) conclusion that the Japanese anaphoric expressions are not compatible with the paycheck strategies.

4.3 Further data and refinement

This subsection discusses slightly more complicated data, laid out in (65) and (66). (65) is an example with a quantifier in each conjunct.¹⁶ Despite the complexity, this sentence is handled by the current definition of a dependency-building conjunction. Therefore, the discussion on (65) shows the generality of the current proposal. Conversely, (66) calls for a slight but straightforward modification of the definition.

- (65) a. (This camp is about fostering a sense of responsibility.)
 This year, [every boy]⁰ was assigned [a cat]¹ and [every girl]² was assigned [a rabbit]³.
 b. Each of them₄ had to take care of it₅ on a daily basis.
- (66) a. [A man]⁰ saw [a donkey]¹, [his₀ son]² saw [a monkey]³, and [his_{0/2} daughter]⁴ saw [a turkey]⁵.
 b. Each of them caught it.

¹⁶I thank an anonymous reviewer of the Third Tsinghua Interdisciplinary Workshop on Logic, Language, and Meaning for bringing this sentence to my attention and emphasizing the importance and the issues it raises for my previous analysis.

(65) is translated as (67), and its update is illustrated in (68). The final concatenation step ($H * (\Delta^{F,G} \cup \Delta^{G,H})$) produces a large table; therefore, the result is omitted to save space. However, it is clear that the information state I grasps all $\text{boy}_n\text{-cat}_n$ and $\text{girl}_n\text{-rabit}_n$ pairs, for which the *being assigned* dependency holds. This dependency is also preserved after concatenation, and the distributive interpretation will resolve the anaphora in (65b).

$$(67) \quad \text{a. } \left(\boxed{\text{A}} \max^0(\text{boy}(0)) \wedge \text{dist}_0(\epsilon_1 \wedge \text{cat}(1) \wedge \text{sg}(1) \wedge \text{be_assigned}(0,1)) \right) \wedge_{\forall} \\ \left(\boxed{\text{B}} \max^2(\text{girl}(2)) \wedge \text{dist}_2(\epsilon_3 \wedge \text{rabbit}(3) \wedge \text{sg}(3) \wedge \text{be_assigned}(2,3)) \right) \\ \text{b. } \text{dist}_4(\text{caught}(4,5) \wedge \text{sg}(5))$$

$$(68) \quad \frac{F \mid \begin{array}{c|ccc} & 0 & 1 & \dots \\ f & \star & \star & \dots \end{array}}{\quad} \xRightarrow{\boxed{\text{A}}} \frac{G \mid \begin{array}{c|cc} & 0 & 1 \\ g_1 & \text{boy}_1 & \text{cat}_1 \\ g_2 & \text{boy}_2 & \text{cat}_2 \\ g_3 & \text{boy}_3 & \text{cat}_3 \end{array}}{\quad}$$

$$\xRightarrow{\boxed{\text{B}}} \frac{H \mid \begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ h_1 & \text{boy}_1 & \text{cat}_1 & \text{girl}_1 & \text{rabbit}_1 \\ h_2 & \text{boy}_2 & \text{cat}_2 & \text{girl}_2 & \text{rabbit}_2 \\ h_3 & \text{boy}_3 & \text{cat}_3 & \text{girl}_3 & \text{rabbit}_3 \end{array}}{\quad}$$

$$\xRightarrow{\Delta^{F,G} \cup \Delta^{G,H}} \frac{I \mid \begin{array}{c|cc} & 1 & 2 \\ i_1 & \text{boy}_1 & \text{cat}_1 \\ i_2 & \text{boy}_2 & \text{cat}_2 \\ i_3 & \text{boy}_3 & \text{cat}_3 \\ i_4 & \text{girl}_1 & \text{rabbit}_1 \\ i_5 & \text{girl}_2 & \text{rabbit}_2 \\ i_6 & \text{girl}_3 & \text{rabbit}_3 \end{array}}{\quad}$$

Consider (66) next. The third conjunct of (66a) is two-way ambiguous in terms of the referent of *his*. The daughter is the grandchild of the man mentioned in the first conjunct if *his* bears index 0, whereas she is the child of that man (i.e., the sister of the boy who saw a monkey) if *his* bears index 2. Both interpretations are felicitous with the intended reading of the anaphora in (66b). The felicity reveals that the dependency-building conjunction should be able to produce the information states G in (69), where the 0th to 5th positions guarantee dynamicity, and the 6th and 7th positions grasp the dependency. The anaphora in (69b), indexed as *Each of them₆ catches it₇*, is resolved through the distributive interpretation with respect to G .

$$(69) \quad \frac{G \mid \begin{array}{c|ccccccccc} & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ g_1 & \text{man} & \text{donkey} & \text{boy} & \text{monkey} & \text{daughter} & \text{turkey} & \text{man} & \text{donkey} \\ g_2 & \text{man} & \text{donkey} & \text{boy} & \text{monkey} & \text{daughter} & \text{turkey} & \text{boy} & \text{monkey} \\ g_3 & \text{man} & \text{donkey} & \text{boy} & \text{monkey} & \text{daughter} & \text{turkey} & \text{daughter} & \text{turkey} \end{array}}{\quad}$$

Nonetheless, multiple occurrences $\wedge_{\forall}$ as in (70) do not provide this result. Let ϕ , ψ , χ be the first, second, and third conjuncts of (66), respectively. Then the information states K , $\Delta^{F,H}$, and $\Delta^{H,K}$ are illustrated as follows. K successfully realizes the internal

and external dynamicity. However, the union of $\Delta^{F,H}$ and $\Delta^{H,K}$ fails to store the person-animal dependency for a future discourse. Intuitively speaking, the pair $\langle \text{daughter}, \text{turkey} \rangle$ is stored in different positions in the information states than the other pairs $\langle \text{man}, \text{donkey} \rangle$ and $\langle \text{boy}, \text{monkey} \rangle$.

$$(70) \quad F[\langle \phi \wedge_{\forall} \psi \rangle \wedge_{\forall} \chi]G \\ \text{iff } \exists H, K : F[\langle \phi \wedge_{\forall} \psi \rangle]H \ \& \ H[\chi]K \ \& \ G = K * (\Delta^{F,H} \cup \Delta^{H,K})$$

$$(71) \quad \begin{array}{c|cccccc} K & 0 & 1 & 2 & 3 & 4 & 5 \\ \hline k_1 & \text{man} & \text{donkey} & \text{boy} & \text{monkey} & \text{daughter} & \text{turkey} \end{array}$$

$$\begin{array}{c|cccccc} \Delta^{F,H} & 0 & 1 & 2 & 3 & 4 & 5 \\ \hline s_1 & \text{man} & \text{donkey} & \text{boy} & \text{monkey} & \text{man} & \text{donkey} \\ s_2 & \text{man} & \text{donkey} & \text{boy} & \text{monkey} & \text{boy} & \text{monkey} \end{array}$$

$$\begin{array}{c|cc} \Delta^{H,K} & 0 & 1 \\ \hline s'_1 & \text{daughter} & \text{turkey} \end{array}$$

Obtaining the desired dependencies for cases of three or more conjuncts thus requires a generalizing of the two-place dependency-building conjunction to the n -place counterpart, defined in (72).

$$(72) \quad \text{Dependency-building conjunction (Dynamic, generalized)} \\ F[\langle \phi_1 \wedge_{\forall} \dots \wedge_{\forall} \phi_n \rangle]G \text{ iff} \\ \exists \vec{H} : F[\langle \phi_1 \rangle]H_1 \ \& \ \dots \ \& \ H_{n-1}[\langle \psi \rangle]H_n \ \& \ G = H_n * (\Delta^{F,H_1} \cup \dots \cup \Delta^{H_{n-1},H_n})$$

The need for this definition suggests that the discourse must be aware of the arity of dependency-building conjunctions. More intuitively, it has to ‘look ahead.’ This should be contrasted with the canonical sequential dynamic conjunction, where a series of conjunctions can be deduced to a consecutive update by each conjunct, as exemplified in (73). Here, the update from F to H_1 by the first conjunct is processed completely ignorant of the incoming discourse. This is not the case for a dependency-building conjunction, which must be aware of the number of conjuncts it has.

$$(73) \quad F[\langle \phi_1 \wedge \phi_2 \wedge \phi_3 \rangle]G \\ \text{iff } \exists H_1, H_2 : F[\langle \phi_1 \rangle]H_1 \ \& \ H_1[\langle \phi_2 \rangle]H_2 \ \& \ H_2[\langle \phi_3 \rangle]G$$

Therefore, I contend that the dependency-building conjunction does not completely replace sequential conjunction. The latter is the default use and dependency-building is performed only when the discourse provides a pragmatic coherence relation that highlights dependency. I return to this matter in Section 5.

4.4 Summary of the proposal

To summarize the proposal: I have argued that the anaphora in (1)/(2) should be analyzed as a special case of quantificational subordination. The observations in Section 2 support this conclusion. To achieve formal implementation of the claim, I propose a new conjunction,

$\wedge_{\forall}$, which preserves the dynamicity of the conjunction and the dependency established in each conjunct.

5 Remaining issues

This section points out two remaining issues. They are both situated in the interplay of semantics and pragmatics: a pragmatic motivation of the distributive operator, and a pragmatic justification of the use of the dependency-building conjunction.

5.1 On covert distributors

Section 2 observed that the felicity of quantificational subordination and SAAC depend on the presence of a quantifier. However, this matter becomes much more complicated by the following observations, in which the anaphoric expressions in the (b)-sentences are subordinate to the dependency established in the (a)-sentences.

- (74) a. [Each candidate]₀ walked to the stage.
b. *He*₀ took [*his*₀ diploma]¹ from the Dean and returned to his seat.
(telescoping; Roberts 1989: 29, attributed to Barbara Partee)
- (75) a. [Every boy]₀ saw [a donkey]₁.
b. Little Johnny gave it₁ a carrot.
(Simon Charlow, p.c.)

A widely accepted analysis of (74), known as *telescoping*, is to posit a covert distributive operator (e.g., Brasoveanu 2007: 247–254), as illustrated in (76).¹⁷

$$(76) \quad \text{dist}_0(\epsilon_1 \wedge \text{diploma_of}(1, 0) \wedge \text{took}(0, 1) \wedge \text{sg}(1))$$

Let G be an information state such that $F\llbracket(74a)G$ for some G . Then $G(0)$ is a set of the candidates, (74) asserts that for every $g \in G$, $g(0)$ took $g(0)$'s own diploma from the Dean.

(75) is a more nuanced case. The reading in question is that the boys saw different donkeys, and little Johnny gave a carrot to the donkey he saw. It differs from the telescoping case in that the sentence is anaphoric to only one of the dependencies established in the preceding context, namely the dependency between Little Johnny and the donkey he saw. This subordination is not predicted by the use of Dist_0 because it universally quantifies over all the dependencies created by (75a), which (at best) results in the hardly available reading in which Little Johnny gave a carrot to all the donkeys seen by the boys.

Thus, one might take the intended reading of (75) as a challenge to the claim that subordination of anaphoras requires the presence of a quantifier. Nevertheless, I conjecture that this anaphora is also resolved by distribution. To illustrate the suggestion, let K be the information state such that $F\llbracket(75a)K$ for some F , visualized as (77).

¹⁷The analysis of possessive pronoun *his* with ϵ here does not have any theoretical significance for the purpose of this paper.

(77)

K	0	1
k_1	boy ₁	donkey ₁
k_2	johnny	donkey ₂
k_3	boy ₂	donkey ₃

The intended reading would result if the distributive operator involved can be more anaphorically specified, so that it only quantifies over $g \in G$ such that $g(0) = \text{johnny}$. Technically, this is achieved by the following definition.

- $F[\llbracket \text{dist}_{n,\text{johnny}}(\phi) \rrbracket G] \text{ iff } F(n) = G(n) \wedge \forall x : \in \{\text{johnny}\} : F|_{n=x}[\llbracket \phi \rrbracket G|_{n=x}]$,
where $F|_{n=x} := \{f \in F : f(n) = x\}$

With this definition in place, the anaphora in question is resolved in the following formula. That is, the sentence asserts that for every $k \in K$ such that $k(0) = \text{johnny}$, it is the case that Johnny gave $g'(1)$ the carrot. Since the information state K preserves the boy-donkey dependency, $k'(1)$ correctly picks up the donkey Johnny saw.

(78) $K[\llbracket \text{dist}_{0,\text{johnny}}(\text{gave_a_carrot_to}(0,1)) \rrbracket G]$

The proposal of course abstracts a few technical assumptions away, such as how to compositionally derive the formula in (78). Still, the claim made here that the necessity of quantifiers to obtain the subordination reading should be interpreted as a theoretical generalization that it necessitates an overt or covert distributive operator. A remaining issue is, of course, when a covert operator is allowed. That the covert **dist** is not freely allowed is seen in the absence of the SAAC reading in (79). The (b)-sentence cannot be interpreted as the man and the son caught the animals they saw.

- (79) a. A man saw a donkey and his son saw a monkey.
b. He caught it.

Furthermore, it was noted above that the usual distributive operator **dist**₀ predicts an unavailable reading for (75) that Johnny gave a carrot to all the donkeys. Why, however, is the reading unavailable while we need to justify the use of the operator in (74)? Unfortunately, this study cannot offer a complete pragmatic theory on this matter, and as far as I am aware, there is not much of this in the literature either. While the use of a covert **Dist** succeeds in description, the theory would lack predictive power without a clear restriction on its use. Such a restriction remains to be found.

5.2 On a pragmatic motivation of the dependency-building

In this study, I formally distinguish the usual sequential conjunction $\wedge$ from the dependency-building conjunction $\wedge_{\forall}$. Furthermore, recall from the discussion around (72) that the generalized dependency-building conjunction must be aware of the arity of the conjunction to produce the desired information states. This means that the interpreter of the sentence must stop processing the conjunction as the usual sequential conjunction $\wedge$ and instead process it as the marked option, $\wedge_{\forall}$. What motivates the interpreter to resort to the marked option then? Although I cannot offer a full-fledged theory on this, I contend that one pragmatic

factor that affects the choice is the presence of *Parallel* relation (Kehler 2002), defined in (80).

(80) **Parallel**

We refer to $p(a_1, a_2, \dots)$, based on the assertion that S_1 and $p(b_1, b_2, \dots)$, based on the assertion that S_2 , where for property vector $\vec{q}$, $q_i(a_i)$ and $q_i(b_i)$ for all i . (Kehler 2002: 16)

The *Parallel* relation is established when the language user recognizes commonalities (or contrasts) between corresponding sets of entities and relations. Suppose that the speaker/hearer identifies a relation $p_1(a_1, \dots, a_n)$ from assertion S_1 and $p_2(b_1, \dots, b_n)$ from S_2 . Then, the *Parallel* result of inferring a common property p , that subsumes p_1 and p_2 , and property vector $\vec{q}$. $\vec{q}$ is a sequence of properties that represent the correspondence of arguments $a_1, \dots, a_n$ and $b_1, \dots, b_n$. For example, the *parallel* relation holds in the pair of sentences in (81). The two predicates (*organized rallies for* and *distributed pamphlets for*) function as parallel relations p_1 and p_2 , which can be subsumed under a more general relation, such as *do something to support*. The subjects (Gephardt and Daschle) serve as parallel arguments a_1 and b_1 , linked by the property q_1 of being prominent Democratic politicians, whereas the objects (Gore and him) serve as parallel arguments a_2 and b_2 , connected by the trivial property q_2 of coreference.

- (81) a. Dick Gephardt organized rallies for Gore.
b. Tom Daschle distributed pamphlets for him. (Kehler (2002): 21)

Kehler (2002) demonstrates that the *Parallel* relation has a linguistic significance. Let us consider the pair in (82). (82b) results from applying *gapping* to (82a). The conjunction in the pre-gapped sentence (82a) allows (at least) two interpretations. In one case, Nan became downright angry *because* Sue became upset. That is, the first conjunct is the cause of the second.¹⁸ In the other interpretation, the events described in the conjuncts happened independently (but, crucially, in a parallel manner). Kehler points out that the gapping example in (82b) allows only the latter interpretation. This contrast leads Kehler to conclude that the establishment of *Parallel* is a necessary condition for gapping.

- (82) a. Sue became upset and Nan became downright angry.
b. Sue became upset and Nan ~~became~~ downright angry.

The SAAC sentences we have seen above all fall in the *Parallel* relation. The representative cases in (1)/(2) have the common property p , namely the seeing relation, between the agents and the animals. The case with three conjuncts in (66) also exhibits the *Parallel* in the same way. The clear *Parallel* relation among the conjuncts motivates the language user to track the dependency within conjuncts, which is implemented through the use of $\wedge_{\forall}$.

Viewed this way, the proposal situates dependency-building conjunctions at the interface of grammar and discourse. The semantics provides the technical possibility of constructing dependency-preserving updates; its actual use is motivated by pragmatic motivations like *Parallel*.

A remaining issue, then, is how strong the *Parallel* relation has to be. For example, the parallelism of the conjuncts in the following English and Japanese examples is less robust

¹⁸For Kehler, the conjunction establishes another coherence relation, namely *Result*.

than the sentences we have observed. While the intended SAAC reading seems to be less accessible in these sentences, the reading is not unavailable either. It might be because there is still a conceivable common property between the conjuncts, like *what the boy and the girl did yesterday*. The Parallel might only predict the robustness of the reading, not its availability. Forming a more predictive pragmatic theory of the distribution of $\wedge_{\forall}$, therefore, has to be left for future work.

- (83) a. A boy opened a window, and a girl bought a bicycle.
 b. Each of them cleaned it afterwards.
 $\rightsquigarrow$?The boy cleaned the window and the girl cleaned the bicycle.

- (84) a. *Otokonoko-ga mado-o ake-te, on'nanoko-ga jitensha-o katta.*
 boy-NOM window-ACC open-AND girl-NOM bicycle-ACC buy.PST
 ‘A boy opened a window, and a girl bought a bicycle.’
 b. *Dochira-mo { sore-o } sooji-sita.*
 Each-all it-ACC clean-did
 ‘Each cleaned it.’
 $\rightsquigarrow$?The boy cleaned the window and the girl cleaned the bicycle.

6 Comparison with other works

6.1 Bumford (2015) on the unification of dynamic conjunction and universal quantification

The use of the dependency-building conjunction in this study provides empirical motivation to take a step further toward the unification of conjunction and universal quantification, which usually does not hold in dynamic semantics. I have argued that dependency-building, which is usually a feature of universal quantifiers, must be augmented to the semantics of conjunction.

The unification is reminiscent of a similar attempt by Bumford (2015), although Bumford argues for unification in the opposite direction. Bumford contends that the dynamicity across conjuncts should be augmented to the semantics of universal quantification. For example, universal quantification in (85a) is analyzed as the dynamic conjunction in (85b).

- (85) a. Every boy saw a donkey.
 b. $\llbracket \text{boy } 1 \text{ saw a donkey} \rrbracket \wedge \llbracket \text{boy } 2 \text{ saw a donkey} \rrbracket \wedge \llbracket \text{boy } 3 \text{ saw a donkey} \rrbracket$

The virtue of this analysis is that it makes universal quantification internally dynamic. The quantification induces sequential update, by which an anaphoric element in the scope of quantification can refer to information consecutively accumulated by the series of conjunctions.

The internal dynamicity, according to Bumford, offers an analysis of the quantifier-internal anaphora, exemplified in (86a) and analyzed as (86b).

- (86) a. Every boy recited a different poem. (Brasoveanu (2011): 94)
 b. $\llbracket \text{John recited a different poem} \rrbracket \wedge$
 $\llbracket \text{Bill recited a different poem} \rrbracket \wedge$
 $\llbracket \text{Fred recited a different poem} \rrbracket$

In the intended reading, (86b) is true if for each pair of different boys $\langle b_1, b_2 \rangle$, there is a pair of different poems $\langle p_1, p_2 \rangle$ such that b_1 recited p_1 and b_2 recited p_2 . The reading is achieved if each conjunct of (86b) introduces a poem that has not been stored in the incoming information states. This line of analysis is implemented by analyzing *a different book* as imposing a condition that the indefinite introduces to the incoming information state a book that is not already stored in the information states, formalized as follows.¹⁹

- $F \llbracket \text{a_different_poem}^n \rrbracket G$ iff $F \llbracket \epsilon_n \wedge \text{book}(n) \wedge \text{sg}(n) \rrbracket G$
 $\& G(n) \notin \{d \mid d = f(m) \text{ for some } f \in F, m\}$

With this definition in place, (86) is translated as (87). Each time `a\_different\_poem` introduces a referent of a poem to an information state, the condition imposed by *different* guarantees that the introduced referent is distinct from any individual in the information states. It thus entails that John, Bill, and Fred each read a different poem, as desired.

- (87) $\epsilon_0 \wedge 0 = \text{John} \wedge \text{a_different_poem}^1 \wedge \text{recited}(0, 1)$
 $\wedge \epsilon_2 \wedge 2 = \text{Bill} \wedge \text{a_different_poem}^3 \wedge \text{recited}(2, 3)$
 $\wedge \epsilon_4 \wedge 4 = \text{Fred} \wedge \text{a_different_poem}^5 \wedge \text{recited}(4, 5)$

Although this paper has argued for taking a step toward unification in the opposite direction, I do not intend to rebut Bumford's direction either. Indeed, we can redefine the Bumford-style universal quantification with $\wedge_{\forall}$, as in (88), without losing the insight of the analysis.

- (88) $\epsilon_0 \wedge 0 = \text{John} \wedge \text{a_different_poem}^1 \wedge \text{recited}(0, 1)$
 $\wedge_{\forall} \epsilon_2 \wedge 2 = \text{Bill} \wedge \text{a_different_poem}^3 \wedge \text{recited}(2, 3)$
 $\wedge_{\forall} \epsilon_4 \wedge 4 = \text{Fred} \wedge \text{a_different_poem}^5 \wedge \text{recited}(4, 5)$

Since the dependency-building conjunction preserves the internal dynamicity, the analysis of `a\_different\_poem` can be intact. A difference between (87) and (88) is that the latter also keeps track of the dependency within the conjuncts. (88) thus produce the inofmration state in (89a), while (87) produces the flat state in (89b).

(89) a.	J	0	1	2	3	4	5	6	7
	j_1	john	poem ₁	bill	poem ₂	fred	poem ₃	john	poem ₁
	j_2	john	poem ₁	bill	poem ₂	fred	poem ₃	bill	poem ₂
	j_3	john	poem ₁	bill	poem ₂	fred	poem ₃	fred	poem ₃
b.	J	0	1	2	3	4	5		
	j_1	john	poem ₁	bill	poem ₂	fred	poem ₃		

¹⁹See Bumford (2015) for a compositional implementation of the analysis.

The dependency preserved in (89a) makes an empirical benefit of facilitating the quantification subordination to quantifier-internal anaphora. For instance, the pronoun *it* in the second clause of (90) is resolved through a by-now familiar distributive interpretation with respect to (89) through **dist** in the second sentence.

(90) Every boy recited a different poem. They each copied it to a note later.

However, the redifition, as well as Bumford’s original proposal, also overgenerates anaphora. It allows a singular pronoun to freely pick up one of the boys as its antecedent in a later discourse. That is, *he* in (91) should be able to refer to Bill given *J* in (90), contrary to fact.

(91) Every boy recited a different poem. #He₂ liked it.

Being aware of this issue, Bumford (2015: 35–36) suggests that the felicity of anaphora is complicated by pragmatic factors beyond what is in principle semantically available. In the case of (91), the universal quantification introduces several equally plausible candidates for the referent of *he*, which hinders pragmatic resolution of the pronoun. However, since this study does not have anything to add to this matter, I leave this issue open.

The discussion in this section thus shows that the dependency-building conjunction can also be incorporated into Bumford’s direction of the unification of conjunction and universal quantification. Furthermore, the use $\wedge_{\forall}$ offers a straightforward analysis of quantificational subordination to quantifier-internal anaphora.

6.2 Brasoveanu (2011) on a pair-based dynamic system

Brasoveanu (2011) proposes a different analysis of the quantifier-internal use of *different*. He argue sthat the universal quantifier literally quantifies over *pairs* of dynamic updates. Using the notation adopted in the current study, the analysis is illustrated as follows. Let the set of relevant boys be $\{\text{john}, \text{bill}, \text{fred}\}$ and F be an empty information state to be updated by the sentence. Then for each pair of different boys x, y , *every* induces a pair of updates of F that correspond to $\langle \llbracket x \text{ recited a poem} \rrbracket, \llbracket y \text{ recited a poem} \rrbracket \rangle$. For example, letting $x = \text{john}$ and $y = \text{fred}$, the pair of updates would be (92a). Note that the two updates in the pair are completely independent from each other, and they both update F separately. Therefore, the updates produce a pair $\langle G_1, G_2 \rangle$ of information states in (92b). At this point, nothing guarantees that poem_1 and poem_2 are distinct. A *different poem* thus imposes a condition that for each such pair of information states that $G_1(1) \neq G_2(1)$.²⁰ The intended reading of (86) is achieved by repeating this process for every pair of different boys.

²⁰The implementation in Brasoveanu (2011) also involves the concatenation of the pair of information states, so that *different* compares two poems in a single information state and imposes distinctiveness. These detail does not concern us here.

(i)

$G * G'$	0	1	2	3
	john	poem ₁	boy ₂	poem ₂

$$(92) \quad \begin{array}{l} \text{a. } \left\langle \begin{array}{l} \epsilon_0 \wedge 0 = \text{john} \wedge \epsilon_1 \wedge \text{poem}(1) \wedge \text{recite}(0,1), \\ \epsilon_0 \wedge 0 = \text{fred} \wedge \epsilon_1 \wedge \text{poem}(1) \wedge \text{recite}(0,1), \end{array} \right\rangle \\ \\ \text{b. } \left\langle \frac{G_1 \mid 0 \quad 1}{\text{john} \quad \text{poem}_1}, \frac{G_2 \mid 0 \quad 1}{\text{fred} \quad \text{poem}_2} \right\rangle \end{array}$$

One may wonder whether a pair-based dynamic system can be employed in the analysis of SAAC. For example, it is conceivable to define a conjunction that explicitly forms a pair of information states.²¹ In this analysis, the conjunction in (1a) induces a pair of (independent) update of an information state, as shown in (93a), which produces a pair of information states in (93b). Presumably, the quantifier *each* in (1b) quantifies over the members of the pair, asserting that for each information state G_n in the pair, $G_n(0)$ catches $G_n(1)$.

$$(93) \quad \begin{array}{l} \text{a. } \left\langle \begin{array}{l} \epsilon_0 \wedge \text{boy}(0) \wedge \epsilon_1 \wedge \text{donkey}(1) \wedge \text{saw}(0,1), \\ \epsilon_0 \wedge \text{girl}(0) \wedge \epsilon_1 \wedge \text{monkey}(1) \wedge \text{saw}(0,1) \end{array} \right\rangle \\ \\ \text{b. } \left\langle \frac{G_1 \mid 0 \quad 1}{\text{boy} \quad \text{donkey}}, \frac{G_2 \mid 0 \quad 1}{\text{girl} \quad \text{monkey}} \right\rangle \end{array}$$

This line of analysis sketched above might be implemented in the following way. First, following Brasoveanu (2011), let sentences denote relations over pairs of information states $\langle F, F' \rangle$. The familiar formulae like predication $R(n_1, \dots, n_m)$, dynamic existential quantification ϵ_n concern only the first item in the pair. That is, they update F and leave F' untouched. The second item becomes only relevant when we have the universal quantification of the kind discussed for quantifier-internal anaphora, or when the conjunction for SAAC is involved. The pair-building conjunction for SAAC, $\wedge_p$, is defined as follows.

- $\langle F, F' \rangle \llbracket R(n_1, \dots, n_m) \rrbracket \langle G, G' \rangle$ iff $F = G \ \& \ F' = G' \ \& \ \langle F(n_1), \dots, F(n_m) \rangle \in I(R)$
- $\langle F, F' \rangle \llbracket \epsilon_n \rrbracket \langle G, G' \rangle$ iff $F' = G' \ \& \ n = |F| \ \& \ \forall f \in F : \exists g \in G : f \rightarrow g \ \& \ \forall g \in G : \exists f \in F : f \rightarrow g$
- $\langle F, F' \rangle \llbracket \phi \wedge_p \psi \rrbracket \langle G, G' \rangle$ iff $\exists H, J : \langle F, F' \rangle \llbracket \phi \rrbracket \langle G, H \rangle \ \& \ \langle F, F' \rangle \llbracket \psi \rrbracket \langle G', J \rangle$

Then a pair-wise distributive operator is defined so that it operates on the incoming pair of information states. It requires that the ϕ updates F to G and F' to G' .

- $\langle F, F' \rangle \llbracket \text{Dist}_p(\phi) \rrbracket \langle G, G' \rangle$ iff $\exists H, J, K : \langle F, F' \rangle \llbracket \phi \rrbracket \langle G, H \rangle \ \& \ \langle F', J \rangle \llbracket \phi \rrbracket \langle G', K \rangle$

Given these definitions, the following formula achieves the intended reading of SAAC.

$$(94) \quad [\epsilon_0 \wedge \text{boy}(0) \wedge \epsilon_1 \wedge \text{donkey}(1) \wedge \text{saw}(0,1)] \wedge_p [\epsilon_0 \wedge \text{girl}(0) \wedge \epsilon_1 \wedge \text{monkey}(1) \wedge \text{saw}(0,1)] \wedge \text{Dist}_p(\text{caught}(0,1))$$

An empirical concern of this analysis is that the conjunction $\wedge_p$ is internally static. As mentioned in the discussion around (52), repeated below, the anaphoricity within the

²¹I thank an anonymous reviewer for *Journal of Semantics* suggesting this interesting possibility.

conjuncts does not hinder the intended reading of SAAC, and thus the internal staticity of $\wedge_p$ is not a welcome feature.

- (95) a. A man saw a donkey and his son saw a monkey.
 b. Each of them caught it.
 $\rightsquigarrow$ The man caught the donkey he saw and his son caught the monkey he (the son) saw.

Furthermore, the pair-building conjunction $\wedge_p$ undergenerates for the cases like (66) repeated below, where more than two conjuncts are involved. Given that the number of conjuncts involved in SAAC is in principle unbound, the analysis of these cases requires lifting the denotation of sentences to be n -tuples of information states, which is a theoretically heavy move.²² Moreover, the full analysis of (96) necessitates the internal dynamicity that is flexible enough so that *his* in the third conjunct can refer either the man in the first conjunct or his son in the second conjunct. The analysis of these anaphora would require establishing anaphoric dependencies *across* the information state in the triple, as in (97). Although implementing the dependency across pairs might not be impossible, it would be conceptually more welcome if the referential dependency in (96) is simply due to the sequential dynamics of the conjunction.

- (96) a. A⁰ man saw a¹ donkey, [his₀ son]² saw a³ monkey, and [his_{0/2} daughter]⁴ saw a⁵ turkey.
 b. Each of them caught it.

$$(97) \quad \left\langle \frac{F}{\text{man}} \mid \frac{0}{\text{man}} \quad \frac{1}{\text{donkey}} , \frac{G}{\text{F(0)'s son}} \mid \frac{0}{\text{F(0)'s son}} \quad \frac{1}{\text{monkey}} , \frac{H}{\text{F(0)/G(0)'s daughter}} \mid \frac{0}{\text{F(0)/G(0)'s daughter}} \quad \frac{1}{\text{turkey}} \right\rangle$$

The analysis based on $\wedge_\forall$ proposed above indeed attributes the dynamics observed in (96) to the sequential dynamics of the conjunction. Furthermore, the proposed analysis does not require significant technical complexity, such as pair-based updates. The only new operation is forward popping, and everything else comes from a standard pluralized dynamic system.

In addition to the conceptual reason for not adopting the pair-based analysis, Brasoveanu himself points out an empirical difference between the quantifier-internal use of *different* and the anaphora seen in quantificational subordination. The former, despite the apparent anaphoricity of *different*, does not exhibit the Weak Crossover effect.²³ Compare (98a), a canonical example of Weak Crossover where *every* cannot bind *his*, with (98b), which allows the reading where for every pair of courses c_1, c_2 , the wine w_1 served in c_1 is different from the wine w_2 served in c_2 .

- (98) a. His mother loves every boy.
 b. A different wine was served with every course.

²²See, however, Bumford and Barker (2013) for a system which takes a *list* of stacks as a model of discourse.

²³The observation at least traces back to Dowty (1985).

In this respect, SAAC and the quantificationally subordinated anaphora, which I have argued should be analyzed in a parallel manner, deviate from *different*. (99) and (100) are clearly degraded in the intended reading.

- (99) a. A boy saw a donkey and a girl saw a monkey.
 b. It scared each of them.
 $\rightsquigarrow^{??}$ For each person x , the animal x saw scared x .
- (100) a. Every student wrote a paper.
 b. It stressed out each of them afterwards.
 $\rightsquigarrow^{??}$ For each person x , the paper x wrote stressed out x afterwards.

A full account of (99)/(100) and the contrast with (98b) awaits a general theory of Weak Crossover.²⁴ Nevertheless, the parallel between (99) and (100) offers further motivation to adopt a similar analysis of these two anaphora, while the deviance between (99a) and (98b) supports a move to analyze the quantifier-internal use differently from them.

7 Conclusion

This paper has investigated *split antecedents across conjunction* (SAAC), in which the antecedents required for resolving a pronoun are distributed across conjuncts. I argued that SAAC is best analyzed in close parallel with quantificational subordination: the quantifier in the second clause triggers a distributive interpretation, and the anaphoric expression itself can remain a simple individual variable.

The main theoretical claim is that the standard sequential dynamic semantics for conjunction is insufficient for SAAC because it does not preserve the dependencies established within each conjunct. To capture the respective interpretation while retaining internal and external dynamicity, I proposed a dependency-building dynamic conjunction $\wedge_{\forall}$. The entry augments the usual sequential update with a dependency-tracking component computed from the transitions between information states. This yields information states that simultaneously (i) achieve the internal and external dynamicity of conjunction and (ii) encode the dependencies needed for SAAC.

Beyond SAAC, the proposal suggests a closer connection between conjunction and universal quantification than is usually assumed in dynamic semantics: dependency-building, typically associated with quantifiers, can also be a property of (some uses of) conjunction. In many the plain static semantics and its extensions such as alternative semantics, conjunction and universal quantification can be inter-defined. Dynamic semantics is peculiar in that it assume they are significantly different. While this paper does not achieve a complete unification of conjunction and universal quantification, it offers a theoretical and empirical motivation to take their similarity more seriously in dynamic frameworks as well.

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²⁴To the best of my knowledge, Weak Crossover with quantificational subordination like (100) has not been explicitly discussed in the literature.

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